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Radix astragali injection enhances recovery from sudden deafness☆

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Abstract

Objectives: An acute interruption of the blood supply to the inner ear is one of the most likely causative factors for sudden deafness (SD). Reactive oxygen species (ROS) have been suggested to be important mediators of the tissue injury during cochlear ischemia and reperfusion. Radix astragali (RA) is natural antioxidant. The aim of this study was to investigate the efficacy of RA in patients with SD.

Patients and methods: We compared the hearing gains from hearing impairment in 46 ears treated with RA with 46 ears treated with non-RA. RA was given intravenously daily for 10 days. There were no significant differences in clinical or audiological data between RA and non-RA groups.

Results: The hearing gain at 250, 500, 1000, 2000, and 4000 Hz in RA group was much higher than that of non-RA group correspondingly ($P < .01$). Also, the hearing gain at PTA (pure-tone average of 250, 500, 1000, 2000, and 4000 Hz) in RA group was significantly higher than that of non-RA group ($P < .01$).

Conclusion: The recovery of hearing was significantly better after treatment of RA than non-treatment of RA. RA can be valuable concurrent therapy for patients with SD.

1. Introduction

Sudden deafness (SD) is considered to be an otologic emergency with no known specific cause. Viral infection, autoimmunity, vascular pathologies, and trauma have been suggested for theoretical causes [1-3]. Among these potential pathogenic factors, an acute interruption of the blood supply to the inner ear is one of the most likely causative factors [4-6], and thus a number of regimens aiming to improve inner ear circulation and oxygen supply have been employed as therapy, including vasodilators, anticoagulants, plasma expanders, carbogen, stellate ganglion block, and hyperbaric oxygen (HBO). Reactive oxygen species (ROS) have been suggested to be important mediators of the tissue injury during cochlear ischemia and reperfusion [7,8], and antioxidants are candidates to alleviate the ischemia-reperfusion injury of the cochlea [7,9]. Also, ROS take an important part in cochlear pathological injury induced by autoimmunity, viral infection, and trauma [10-13].

Radix Astragali (RA) is a kind of traditional Chinese herb, which is widely used in treating ROS-mediated injury in various organs through its anti-oxidant properties [14-18]. RA has been used for treatment of deafness for hundreds of years in Chinese Medicine history. From these, we have speculated RA has the potential effects for treatment of SD. From 2005, RA has been a part of the routine treatment of SD in our department. In this study we investigate the effect of RA therapy on the recovery from SD.

☆ Declaration of interest: The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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2. Materials and methods

2.1. Patients and evaluation

The study material consisted of two patient groups: RA group and non-RA group. All patients of SD adopted in this study were treated in the Department of Otolaryngology, Guangzhou General Hospital of Guangzhou Military Command from September 1998 to August 2011. All patients met the following inclusion criteria: (1) sudden onset of sensorineural hearing loss; (2) unknown cause of hearing loss; (3) nonfluctuating hearing loss; (4) A hearing loss of 30 dB or greater in at least 3 contiguous frequencies formed the audiometric inclusion criterion. (5) time from the onset of hearing loss to the start of treatment ≤5 days. The patients underwent standard otolaryngologic and otoneurologic examination after admittance to the hospital. All patients were evaluated with an audiometric test battery consisting of pure tone audiometry, tympanometry, and auditory brainstem evoked responses. The severity of hearing loss was assessed by using the average hearing level at frequencies of 250, 500, 1000, 2000, and 4000 Hz, as measured at the first visit. If a patient did not respond to the maximum level of sound generated by the audiometer, 5 dB was added to the maximum level to determine a threshold value. Unless the patient refused, magnetic resonance imaging (MRI) of the internal auditory canal/cerebellopontine angle was performed during or after treatment, and acoustic tumors and brainstem lesions were thus ruled out. The patients were observed at the hospital ward and treated with RA or non-RA. The study was approved by the Ethics Committee of Guangzhou General Hospital of Guangzhou Military Command.

2.2. Patient groups

2.2.1. RA group

RA group consisted of 46 patients (46 ears) treated with RA during a period from May, 2005 to August, 2011. RA (Chengdu Diao Pharmaceutical Company, Sichuan, China; was approved by Chinese Drug Administration, approved Number:2001ZFB0171.) was administrated intravenously at the dosage of 0.5 ml/kg/d for 10 continuous days. 10 ml injection of RA contains 20 g of crude medication astragalus membranaceus.

2.2.2. Non-RA group

For each RA treated patient a best matching non-RA control was selected. The control group of 46 patients (46 ears) was selected from the total of 221 patients with SD treated with non-RA from September, 1998 to May, 2004. The adopted criteria were: (1) the same gender; (2) very near age (within 3 years old); (3) almost equal delay of time from the onset of hearing loss to the start of treatment (limit of 12 hours); (4) the same feature of tinnitus; (5) the same characteristic of vertigo; and (6) similar amount and audiogram configuration of the initial hearing impairment (pure-tone average of 250, 500, 1000, 2000, and 4000 Hz (PTA), maximal hearing loss (max HL) with a difference ≤5 dB). The matching was carried out without any information of the patients' outcome.

2.3. Measurements of hearing

The first hearing measurement was done as soon as possible after arrival at our department and before any treatment. A qualified audiometric technician measured the pure-tone thresholds in a sound proof chamber using a clinical audiometer (GSI 61 clinical audiometer) calibrated by national sound center every year. Then tympanometry and auditory brainstem evoked responses were detected. The follow-up audiograms were taken at the end of the therapy (on day 10) and the last measurement in 12 weeks if some degree of damage was present on day 10. The hearing level at 12 weeks after the initiation of treatment was considered to be fixed. Pure-tone audiograms were compared using the averaged hearing loss in dB at 250, 500, 1000, 2000, and 4000 Hz (PTA). Absolute hearing gain was defined as the difference in hearing thresholds in decibels before and after the treatment.

2.4. Treatment

All patients were strictly instructed to not smoke or drink caffeinated beverages during the 10 days of hospitalization. Mental fatigue was suggested to be avoided, doing slightly sports were encouraged simultaneously. Both RA group and non-RA group were treated with dexamethasone intravenously at the dosage of 0.2 mg/kg/d for 10 successive days, and hyperbaric oxygen therapy daily. Compound vitamin B was administrated 2 tablets three times a day. Salvia injection which is a vasodilator was also administrated. RA group received the same basic treatment as non-RA group, with the addition of RA therapy.

2.5. Statistics

Data values are expressed as means ± SEM. Statistical differences of ratios and means were analyzed using the x^2 test and t test, respectively. Differences were considered to be significant when P was < .05.

3. Results

3.1. Profile of patients

Table 1 shows the profiles of the patients in RA and non-RA groups. There were no significant differences in age, association with vertigo and tinnitus, and interval between onset and treatment between the two groups. Also, there were no significant differences in the initial hearing levels between RA and non-RA groups (P > .05).

3.2. Therapeutic outcomes

The overall therapeutic outcomes are summarized in Table 2. The average hearing gain of PTA was 36.2 ± 9.4 dB and 21.1 ± 8.2 dB in RA and non-RA groups, respectively. It

Table 1
Profiles of patients

	RA group	non-RA group	*P*
No. of patients	46	46	
Age (years)			
Mean ± SEM	45.7 ± 6.9	44.2 ± 7.1	>.05[a]
Range	34-66	33-65	
Vertigo	3	3	>.05[b]
Tinnitus	40	40	>.05[b]
Initial hearing level (dB)			
Mean ± SEM(PTA)	68.3 ± 9.5	65.9 ± 7.4	>.05[a]
Range	55-85	55-85	
Interval between onset and treatment (days)			
Mean ± SEM	3.9 ± 2.7	3.6 ± 2.2	>.05[a]
Range	1-5	1-5	

PTA: pure-tone average of 250, 500, 1000, 2000, and 4000 Hz.
[a] t test.
[b] x^2 test.

demonstrated the hearing gain in RA group was much higher than that of non-RA group ($P < .01$). Also, the hearing gain at 250, 500, 1000, 2000, and 4000 Hz in RA group was much higher than that of non-RA group correspondingly ($P < .01$).

4. Discussion

SD is not a disease per se, it is considered as a manifestation of an underlying cochlear pathology. No therapy produces specially effects on SD at this moment. It is general consensus that steroids are effective with route of administration by oral, intravenous and intratympanic [19-21]. Most authors considered intratympanic steroids to be more effective, some recommended it as salvage treatment [21]. At least, the transtympanic route presents two main advantages: i) presenting a higher concentration of the drug in the cochlea and, ii) reducing the side-effects due to systemic absorption. Steroid therapy in combination with complex treatment regimens produced better effect than steroid therapy alone, such as steroid therapy combination with hyperbaric oxygen for SD [22]. ROS takes important part in the pathogenesis of sudden hearing loss. Antioxidants have been used both systematically and transtympanically [23,24]. Comprehensive treatments have been used in our department for decades. This study investigated the efficacy of RA therapy in SD by comparing the hearing gains in 46 ears of 46 patients given RA and matching 46 ears of 46 patients given non-RA. The hearing gains at 250, 500, 1000, 2000, 4000 Hz and PTA in RA group were significant higher than those in non-RA group ($P < .01$). The results of this study demonstrate that RA is useful for patients with SD.

Table 2
The hearing gains in RA and non-RA group (n = 46, dB HL)

	RA group	non-RA group
250Hz	46.2 ± 12.4	30.7 ± 9.1[a]
500Hz	35.4 ± 8.3	20.1 ± 6.7[a]
1000Hz	35.6 ± 10.3	21.9 ± 7.9[a]
2000Hz	32.5 ± 11.8	18.4 ± 6.1[a]
4000Hz	30.1 ± 8.8	17.8 ± 7.3[a]
PTA	36.2 ± 9.4	21.1 ± 8.2[a]

PTA: pure-tone average of 250, 500, 1000, 2000, and 4000 Hz.
[a] RA group compared with non-RA group, $P < .01$ (t test).

Many authors have suggested that vascular disorder could be the main cause of SD. Patients with systemic vascular or hematologic abnormalities have a higher risk of suffering from SD [25]. SD could be induced in the patient with malignant hypertension after a rapid reduction in blood pressure [26]. Thus, it demonstrated SD could be caused by ischemia. Sudden vestibulocochlear nerve deficit may be the only manifestation in the patient with anterior inferior cerebellar artery infarction [27]. Furthermore, SD could be a characteristic prodromal symptom of a cerebrovascular disturbance [28,29].

The animal model of SD induced by artificial ischemia of the inner ear has been made successfully [30,31]. In sudden occlusion of the major vascular supply to the inner ear of guinea pigs, the cochlea was found to be more vulnerable than the vestibular labyrinth; the outer and inner hair cells and stria vascularis were most often affected [32].

Reoxygenation after ischemia-reperfusion causes the accumulation of superoxide anions in cochlea. The superoxide anion is converted by superoxide dismutase to hydrogen peroxide, and hydrogen peroxide react with iron to produce hydroxyl radical, which is a potent oxidant capable of promoting oxidative damage. Hydroxyl radicals can oxidize DNA, proteins, and plasma membrane [33]. The iron concentration in the perilymph of the cochlea was increased when animal was hypoxia [34], and much increased production of hydroxyl radicals was detected during the re-oxygenation phase after hypoxia [8]. These demonstrated that hydroxyl radicals were produced during the early reperfusion period in cochlea. Nitric oxide (NO) also takes part in the mechanism of cochlear ischemia-reperfusion injury. A significant increase of NO production in the perilymph was detected in the reperfusion period after transient ischemia [35]. NO generated by neuronal NOS (nNOS) and inducible NOS (iNOS) deteriorates the cochlea in ischemia reperfusion injury [36-38]. And iNOS inhibitor decreased the postischemic shift of the ABR or CAP threshold [36,38]. These results definitely demonstrate that NO generated by nNOS and iNOS damages cochlea after reperfusion.

Additionally, autoimmunity, viral infection, and trauma damage cochlea through the ROS formation [10-13]. It is general consensus that autoimmunity, viral infection, trauma are possible causes of SD [1-3].

It has been demonstrated that ROS had been reduced by RA in myocardial ischemia reperfusion in rats [39]. RA is effective in reversal of left ventricular remodeling and improving left ventricular function in patients with acute myocardial infarction through its antioxidant properties [17]. Also, RA can effectively reduce cisplatin ototoxicity through the inhibition of apoptosis of hair cells [40]. The results of this study demonstrated RA is useful for sudden deafness, and the possible mechanism may be its antioxidant property. RA has been widely used in China for thousand of years, and no harmful side effects have been reported.

4.1. Limitations of this study

In theory, the study should be better to have randomized treatment in a prospective manner to include a group that received standard therapy plus RA and a group that received standard therapy plus placebo. Additionally, although the results are compared with regard to decibel improvement both at individual frequencies and on a five frequency PTA over defined period of observation, there is no certainty that the baseline hearing in these two groups are similar. This would possibly pose a “ceiling effect” in terms of degree of improvement, and it may be a potential confounding variable.

5. Conclusion

This set of studies demonstrates that the recovery of hearing in SD was significantly better after treatment of RA than non-treatment of RA. RA can be valuable concurrent therapy for patients with SD.

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References

[1] Yamasoba T, Kikuchi S, Higo R, et al. Sudden sensorineural hearing loss associated with slow blood flow of the vertebrobasilar system. Ann Otol Rhinol Laryngol 1993;102:873-7.
[2] Hughes GB, Freedman MA, Haberkamp TJ, et al. Sudden sensorineural hearing loss. Otolaryngol Clin North Am 1996;29:393-405.
[3] Pullen FW, Rosenberg GJ, Cabeza CH. Sudden hearing loss in divers and fliers. Laryngoscope 1979;89(9 Pt 1):1373-7.
[4] Cole RR, Jahrsdoerfer RA. Sudden hearing loss: an update. Am J Otol 1988;9:211-5.
[5] Gussen R. Sudden deafness of vascular origin: a human temporal bone study. Ann Otol Rhinol Laryngol 1976;85(1 Pt 1):94-100.
[6] Kim JS, Lopez I, DiPatre PL, et al. Internal auditory artery infarction: clinicopathologic correlation. Neurology 1999;52:40-4.
[7] Seidman MD, Quirk WS, Nuttall AL, et al. The protective effects of allopurinol and superoxide dismutase-polyethylene glycol on ischemic and reperfusion-induced cochlear damage. Otolaryngol Head Neck Surg 1991;105:457-63.
[8] Hara A, Serizawa F, Tabuchi K, et al. Hydroxyl radical formation in the perilymph of asphyxic guinea pig. Hear Res 2000;143:110-4.
[9] Seidman MD, Quirk WS. The protective effects of tirilated mesylate (U74006F) on ischemic and reperfusion-induced cochlear damage. Otolaryngol Head Neck Surg 1991;105:511-6.
[10] Watanabe K, Tomiyama S, Jinnouchi K, et al. Expression of inducible nitric oxide synthase in the cochlea following immune response in the endolymphatic sac of guinea pigs. ORL J Otorhinolaryngol Relat Spec 2001;63:155-9.
[11] Schachtele SJ, Mutnal MB, Schleiss MR, et al. Cytomegalovirus-induced sensorineural hearing loss with persistent cochlear inflammation in neonatal mice. J Neurovirol 2011;17:201-11.
[12] Xiong M, He Q, Lai H, et al. Oxidative stress in spiral ganglion cells of pigmented and albino guinea pigs exposed to impulse noise. Acta Otolaryngol 2011;131:914-20.
[13] Kim SJ, Park C, Han AL, et al. Ebselen attenuates cisplatin-induced ROS generation through Nrf2 activation in auditory cells. Hear Res 2009;251:70-82.
[14] Chan JY, Lam FC, Leung PC, et al. Antihyperglycemic and antioxidative effects of a herbal formulation of Radix Astragali, Radix Codonopsis and Cortex Lycii in a mouse model of type 2 diabetes mellitus. Phytother Res 2009;23:658-65.
[15] Chan JY, Leung PC, Che CT, et al. Protective effects of an herbal formulation of Radix Astragali, Radix Codonopsis and Cortex Lycii on streptozotocin-induced apoptosis in pancreatic beta-cells: an implication for its treatment of diabetes mellitus. Phytother Res 2008;22:190-6.
[16] Huang X, Liu Y, Song F, et al. Studies on principal components and antioxidant activity of different Radix Astragali samples using high-performance liquid chromatography/electrospray ionization multiple-stage tandem mass spectrometry. Talanta 2009;78:1090-101.
[17] Zhang JG, Gao DS, Wei GH. Clinical study on effect of Astragalus injection on left ventricular remodeling and left ventricular function in patients with acute myocardial infarction. Zhongguo Zhong Xi Yi Jie He Za Zhi 2002;22:346-8.
[18] Wang D, Shen W, Tian Y, et al. Protective effect of active components extracted from radix Astragali on human erythrocyte membrane damages caused by reactive oxygen species. Zhongguo Zhong Yao Za Zhi 1996;21:746-8.
[19] Chen CY, Halpin C, Rauch SD. Oral steroid treatment of sudden sensorineural hearing loss: a ten year retrospective analysis. Otol Neurotol 2003;24:728-33.
[20] Wilson WR, Byl FM, Laird N. The efficacy of steroids in the treatment of idiopathic sudden hearing loss. A doubleblind clinical study. Arch Otolaryngol 1980;106:772-6.
[21] Ho HG, Lin HC, Shu MT, et al. Effectiveness of intratympanic dexamethasone injection in sudden-deafness patients as salvage treatment. Laryngoscope 2004;114:1184-9.
[22] Fujimura T, Suzuki H, Shiomori T, et al. Hyperbaric oxygen and steroid therapy for idiopathic sudden sensorineural hearing loss. Eur Arch Otorhinolaryngol 2007;264:861-6.
[23] Joachims HZ, Segal J, Golz A, et al. Antioxidants in treatment of idiopathic hearing loss. Otol Neurotol 2003;24:572-5.
[24] Korver KD, Rybak LP, Whitworth C, et al. Round window application of D-methionine provides complete cisplatin otoprotection. Otolaryngol Head Neck Surg 2002;126:683-9.
[25] Schweinfurth JM, Parnes SM, Very M. Current concepts in the diagnosis and treatment of sudden sensorineural hearing loss. Eur Arch Otorhinolaryngol 1996;253:117-21.
[26] Chao TK. Sudden sensorineural hearing loss after rapid reduction of blood pressure in malignant hypertension. Ann Otol Rhinol Laryngol 2004;113:73-5.
[27] Lee H, Ahn BH, Baloh RW. Sudden deafness with vertigo as a sole manifestation of anterior inferior cerebellar artery infarction. J Neurol Sci 2004;222:105-7.
[28] Mun SK, Hong YH, Kang SH, et al. A case of temporal intracerebral hemorrhage that presented with sudden bilateral hearing loss as the initial symptom. J Korean Neurosurg Soc 2010;48:438-40.
[29] Davidi ES, Warman M, Lahav Y, et al. Cerebellar ischemia initially presented as sudden sensorineural hearing loss and gaze-evoked nystagmus. Isr Med Assoc J 2010;12:55-6.
[30] Asai Y, Umemura K, Kohno Y, et al. An animal model for hearing disturbance due to inner ear ischemia: photochemically induced thrombotic occlusion of the rat anterior inferior cerebellar artery. Eur Arch Otorhinolaryngol 1993;250:292-6.
[31] Schweinfurth JM, Cacace AT. Cochlear ischemia induced by circulating iron particles under magnetic control: an animal model for sudden hearing loss. Am J Otol 2000;21:636-40.
[32] Kimura RS. Animal models of inner ear vascular disturbances. Am J Otolaryngol 1986;7:130-9.

[33] Samokyszyn VM, Aust SD. Iron. In: Zelenock GB, D'Alecy LG, Fantone III JC, Shlafer M, & Stanley JC, editors. Clinical ischemic syndromes. St. Louis, MO: Mosby Co.; 1990. p. 63-88.
[34] Hara A, Komeno M, Senarita M, et al. Effect of asphyxia on the composition of cationic elements in the perilymph. Hear Res 1995;90:228-31.
[35] Morizane I, Hakuba N, Hyodo J, et al. Ischemic damage increases nitric oxide production via inducible nitric oxide synthase in the cochlea. Neurosci Lett 2005;391:62-7.
[36] Tsuji S, Tabuchi K, Hara A, et al. Long-term observations on the reversibility of cochlear dysfunction after transient ischemia. Hear Res 2002;166:72-81.
[37] Tabuchi K, Ito Z, Wada T, et al. Effect of 7-nitroindazole upon cochlear dysfunction induced by transient local anoxia. Ann Otol Rhinol Laryngol 2000;109(8 Pt 1):715-9.
[38] Tabuchi K, Tsuji S, Asaka Y, et al. Ischemia-reperfusion injury of the cochlea: effects of an iron chelator and nitric oxide synthase inhibitors. Hear Res 2001;160:31-6.
[39] Ying M, Li X, Sun C, et al. Effect of total flavonoids of radix Astragali on free radical production by myocardial ischemia-reperfusion in rats. Zhongguo Zhong Yao Za Zhi 1996;21:304-5.
[40] Tian Y, Yu G, Jiang X. The Protective Role of Radix Astragali against Ototoxicity Caused by Cisplatin. Journal of Audiology and Speech Pathology 2009;17:261-4.